
MANUSCRIPT REVIEW · SYNTHESIZED LEDGER

Complete Line-Level Review Ledger of the Theory Companion v2

The full record of two sequential line-level passes over companion_categorical_v2.tex (4,444 lines): nineteen verified repairs, twenty-four persisting flaws, seven repair-induced issues, eighteen new findings, the severity triage, the verified-sound list, and the structural line map - every point from both passes, merged without loss.

Editorial Assessment Unit

Two sequential reads of all 4,444 lines · hand re-derivations of the suspect computations · grep-verified singleton occurrences · arithmetic spot-checks of every load-bearing number

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1. Scope, Sources, and Verification Method

This document is the complete, synthesized line-level review ledger of the theory companion paper **companion_categorical_v2.tex** (4,444 lines). It merges the two prior reading ledgers - the first sequential read (saved as `research/companion_v2_read_findings.md`) and the focused re-read for flaws and internal inconsistencies (`research/companion_v2_reread_findings.md`) - into a single reference deliverable. The synthesis rule is strict: **every valid point from both ledgers appears here, none is dropped, and none is condensed away**. Where the two passes describe the same defect at different sharpness levels, the entries are merged with the sharpened wording preserved; where a point appears in only one pass, it is carried over in full.

The file under review is the standalone theory paper "Stratified Connections, Optic Composition, and the Homotopy Fixed-Point Extension" (author X, September 2026). It positions itself as the categorical and homotopy-theoretic companion to an application paper cited under the key `zai2026measure` (the metabolic-curvature-measure repository). Its header (lines 1-52) declares that it implements a set of pending mathematical repairs: the repaired envelope chain, a terminal-coalgebra treatment of `maxRAF` via a deflationary operator, the demotion of functorial realization to a conjecture, the pairs-crossing small-loop expansion, and the boundary reset at its true order. Its four layers are: (i) the SAVGS object and the 2-category `StCon(B)` of stratified connections; (ii) a seven-optic composition theorem with a contraction analysis; (iii) the filtered-colimit construction of autocatalytic (RAF) sets; and (iv) a homotopy-type-theoretic extension with explicitly marked proof-sketch status.

Two passes produced the raw findings. The **first pass** read all 4,443 lines sequentially and logged 19 verified repairs against the earlier SAVGS manuscript (the 131-page `v3`), 22 persisting flaws, and 7 issues newly introduced by the partial repairs; it also recorded that the implemented repairs correspond one-to-one to items in the 2026-09-02 math-coherence analysis (items `T1/T2-partial`, `D4`, `T4-as-posetal`, `T6-partial`, and the `C_fat` derivation). The **re-read** then re-traversed all 4,444 lines with a fresh focus on internal inconsistencies: it re-verified all 22 persisting flaws and the 7 `v2`-specific issues, re-derived the suspect computations by hand, sharpened several flaws, and added 18 new findings (`N1-N18`) together with a severity triage and an eight-point minimal repair set.

Verification methods used across the two passes: sequential line-level reading with line-number bookkeeping; hand re-derivation of suspect computations (the Gaussian variance $\text{Var}(G) = (5/4)sa^2$, the free-Brownian-motion Lévy-area variance, the closure degeneracy $\text{cl}(F \cup \text{prod}(S)) = F \cup \text{prod}(S)$, the composite range escape of the seven-map chain, the 0.845-versus-0.8485 numeric mismatch); `grep` verification of singleton occurrences (the phrase "42/42" occurs exactly once; the symbol `a*` is defined nowhere; the "residual-compatibility condition" is stated nowhere); and arithmetic spot-checks of every load-bearing number ($0.92^6 \cdot 1.15 = 0.6973$; the triangle, pentagon, circle, and ellipse loop areas; the Table `n4` ratios 29 and 56; $F1 = 0.8625$; $375 = 5 \cdot 5 \cdot 3 \cdot 5$ and $152 + 219 + 4 = 375$; the extended-network RAF check reaction by reaction; and the appendix terminal-coalgebra proof end to end).

4,444 lines read in both passes	19 verified repairs vs the v3	24 persisting flaws (merged)	18 new re-read findings
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2. Verified Repairs Against the Earlier Manuscript (R1-R19)

These nineteen repairs were verified at line level in the first pass and re-confirmed in the re-read. They are the genuine improvements of the theory paper over the SAVGS v3 manuscript, and each is stated with its line evidence. Cross-references in brackets point to related persisting flaws or new findings recorded in chapters 3-5, where the repair is incomplete or where the re-read sharpened the picture. The companion's repairs implement, item by item, the 2026-09-02 math-coherence plan: the pairs-crossing holonomy repair (item T1/T2-partial), the softplus envelope chain (T4), the deflationary terminal coalgebra (T4-as-posetal), the exact C_{fat} derivation, the realization demotion (T6-partial), and the countable-dense fix for the smooth-envelope supremum.

R1 - Vereshchagin-Vitányi attribution now present (L3039).

The remark `rem:envelope-distD` now cites `vereshchagin2010rate`: the algorithmic rate-distortion distance R_L converges to dist_D up to the standard additive constant along the code enumeration. Categorical cybernetics is also now cited (`hedges2024cybernetics` at L3467, `fong2017backprop` at L3536), and the reference list carries 44 unique entries. The v3's uncited-prior-art exposure on its central algorithmic tool is closed for this paper. [The adjacent prior art on closure combinatorics remains uncited - see F22.]

R2 - Pairs-crossing small-loop repair (L683-706, `Thm thm:stratified-holonomy`).

A closed loop now crosses a constraint-switching wall an even number of times; the boundary term is the tangential variation $\log g(p+) - \log g(p-) = O(\epsilon)$, and an $O(1)$ reset contribution is claimed only for open arcs. The proof (L709-739) shows the exact cancellation in the abelian case. The numerics (remark on the gluing verification, 10^{-13} agreement, $n_{\text{cross}} = 2$) match the regime that was independently verified in the v3 read. The v3's $O(1)$ -per-crossing error is gone. [The remainder order label is still wrong in general - see V2.]

R3 - The matching condition (O3) repaired to the standard gauge transform (L576-592).

The definition now reads $A_j = g^{-1} A_i g + g^{-1} dg$ in matrix form, with the infinitesimal form and the abelian reduction $A_j = A_i + d(\log g)$ stated correctly. [The theorem hypothesis still states the old garbled form - the internal mismatch is V1, and its one-line fix is drafted in the companion repair guide.]

R4 - The C^2 falsehood retracted (Def `ard-surrogate`, L2830-2850).

The surrogate kernel is now a softplus ψ_s replacing the squared positive part $[d - D]_+^2$, and the paper states explicitly that the smoothness of the kernel, not of the positive part, is what carries the C^2 claim. The proposition's proof (L2864-2876) adds: "What is NOT claimed is C^2 regularity of the positive part or of suprema: those live at the envelope level, where the correct regularity concept is the Clarke subdifferential." The v3's false C^2 claim is fully retracted.

R5 - Lemma uniform-lip repaired (L2908-2944).

The parameter box is now compact, $P = [\tau_{\min}, \tau_{\max}] \times [\beta_{\min}, \beta_{\max}] \times [0, D_{\max}] \times [s_{\min}, s_{\max}]$, with a uniform Lipschitz constant $L \leq L_{\max}$, and the denominator is bounded below by the positive quantity $2^{-L_{\max}/\tau_{\min}} e^{-\beta_{\max}(R_{\max} + s_{\max} \log 2)^2/\tau_{\min}}$. The v3's vacuous limit bound (the bound tending to 0 as the parameters shrink) is gone. [A numerator gap remains in the written proof - see V4.]

R6 - Countable dense subfamily for the envelope supremum (Def smooth-envelope, L2892-2906).

The supremum over the compact parameter box equals the supremum over any countable dense subset Q , by continuity; this fixes the v3's problem of taking a supremum over a continuum while claiming enumerability (the v3's theorem 4.11(e) issue).

R7 - Envelope regularity stated honestly (Thm smooth-envelope, L2946-3027).

The theorem claims: (a) Lipschitz continuity; (b) Danskin/Milgrom-Segal differentiability at singleton argmax, with an explicit sentence "No C^2 claim is made at the envelope level"; (c) the Clarke subdifferential as the closed convex hull of active gradients; (e) effective approximability by an ϵ -net. [Item (e)'s domination claim has two defects - see F6.]

R8 - Corollary 4.14 repaired into a declared computation (Cor kappa-geometric-phase, L3065-3117).

The corollary is now titled "Geometric-phase computation ...; model-specific" and declares the calibrated connection $A = ((V - V_{\max})/2V_{\max})d\theta$. Item (iii) states that this is a computation for the radially symmetric model with the calibrated connection, not an identity between D_V and κ_V . The proof was verified in the re-read: the flux computation gives the loop integral of A as $\pi\delta V/V_{\max}$, and the quadratic case $\pi a^2/r^2$ was re-derived and checks. The v3's incoherent proof (computing πa^4 while claiming πa^2) is repaired. [A sign slip remains in the displayed formula - see N7.]

R9 - The 3/2 fatigue constant derived exactly (Lemma ito-expand L2278-2344 + Thm levy-3half L2346-2382).

The Gaussian variance $\text{Var}(G) = (5/4)\sigma^2 a^2$ is derived by Ito isometry ($\sigma^2 a^2/4$) plus drift ($\sigma^2 a^2/2$, with the cosine and sine loop variances $V_c = 1/8\pi^2$ and $V_s = 3/8\pi^2$, both verified) plus cross-covariance ($\sigma^2 a^2/4$, verified), giving $C_{\text{fat}} = \sqrt{(5v)}/2$. The v3's " κ of order unity" fudge is gone. The Monte Carlo two-term fit gives $c_1 = 0.349$ against the exact 0.3536 (1.5%), and the confidence interval [1.471, 1.493] excluding 3/2 is now explained by the sub-leading analytic term dragging the single-exponent fit. All numbers check. [The Lévy-area term's normalization is inconsistent three ways - see N1; and the mean/fluctuation conflation is N2.]

R10 - Terminal-coalgebra theorem repaired and proved (Def deflationary L3469-3490; Thm maxraf-terminal L3492-3515; Appendix proof L4330-4400).

The operator $D(S) = S \cap \Phi(S)$ is deflationary by construction, fixing the v3's false inclusion $\Phi(S) \subseteq S$. The terminal coalgebra is obtained posetally via Tarski (greatest post-fixed point equals greatest fixed point - correct), the Hordijk-Steel iterative removal equals $D^n(U)$ step for step, the iteration terminates in at most $|U|$ steps, and the procedure is polynomial. The remark on the community result (L3528-3543) explicitly disclaims the v3's Set-endofunctor overclaim: "the lattice-level statement above is the one that carries a complete proof." [The catalysis-condition mismatch remains - see V5.]

R11 - Theorem B demoted to a conjecture (Conj functorial-realization, L3546-3560, with remark L3562-3577).

The uniqueness clause would require a parametricity argument "that we likewise do not supply; the statement is therefore recorded as a conjecture rather than a theorem." The bogus Strachey-Reynolds proof of the v3 is retracted. The app-paper cross-check (companion deliverable) shows the application paper no longer relies on the functorial claim.

R12 - Realization defined at the set level (Def realization, L1723-1745).

$R(O)(x) = \text{bwd}_1(\text{Res}_O, \text{fwd}_n \cdots \text{fwd}_1(x))$ is now an explicit set-level definition, with the honest sentence "Whether R extends to a genuine functor ... is an open problem." The v3's undefined "realization functor R " is repaired. [The Lipschitz analysis still bounds only the forward chain - see F3.]

R13 - Colimit/limit confusion fixed in place (Con invlim L3140-3149; Prop invlim-extended item 2, L3376-3381).

The construction is now correctly labeled "This is a filtered colimit, not an inverse limit," and the 15 structure arrows $R_i \rightarrow R_{\max}$ are described as inclusions witnessing the universal property of the colimit (a cocone), replacing the v3's backwards "projection arrows witnessing the limit."

R14 - The $0 = 0$ "falsifiable" claim retracted (Remark zero-zero, L3398-3413).

The paper now states: "a consistency check of the implementation, not an independent falsifiable prediction ... both computed quantities vanish structurally." Falsifiability is moved to items 1-2 (the directed-system axioms do fail generically).

R15 - The network battery honestly reframed (Sec network-battery L3983-3999 + Remark L4249-4261).

The battery is now presented as a "designed progression ... measures the test's sensitivity" rather than an empirical discovery, with a fixed-model arm on the unmodified iJO1366. The v3's "STRICTLY GREATER" proposition-dressing of design iterations is replaced by a summary table. The protocol is single ($T = 200$, $\delta = 0.05$, $kcat = 0.5$), collapsing the v3's $T \in \{200, 300, 500\}$ goalposts. [The G/H row contradiction persists - see N3; the dropped restoration control is N4.]

R16 - Noether repaired (Prop noether, L1294-1331).

The hypotheses are now the correct ones (the Hessian-isometry condition $L_{\xi g} \varphi = 0$ plus $\xi(U) = 0$), with the affine-Bregman corollary following, and a remark (L1366-1377) stating: "not divergence invariance alone (which is false in general ...), but affine isometries of the Hessian metric." The v3's false claim is self-corrected. [The Noether-current display is still garbled - see N8.]

R17 - Provenance and voice clean (L4402-4438).

No "the user," no Qwen/audit references, no commit hashes, no version numbers (v10/v11 gone), no build logs, no LaTeX patch narratives. The declarations carry an AI-use disclosure (GLM and DeepSeek, under the author's direction), a sole-author statement, and data/code availability with SHA-256 manifests. The entire integrity-signal layer that plagued the v3 is gone. [The reference file `journal_manuscript_refs.bib` is not part of the upload, so reference resolution cannot be checked - see F24.]

R18 - Open problems section honest (L4265-4323).

Seven genuine open problems are listed, including the boundary term at $O(\epsilon^2)$, endofunctor semantics, filtered colimits beyond Set, the explicit ∞ -optic composition ("the most significant gap"), sharpening the contraction bound, third-party benchmarks, and the Phase III strictness account.

R19 - Smaller repairs.

The hierarchy definition now cites def:kv correctly (the v3 cited "Definition 2.1" wrongly); the Claim-G $R^2 = 1.0000$ mechanism is honestly explained (L2581-2594); the Liouvillian gap versus energy variance is disambiguated (Def lindbladian, L2445-2463); the Zeno contraction proof is written out cleanly and correctly, with $\Psi(\rho) = \mu\rho + (1 - \mu)P/k$ and $\mu = (1 - p)/[(1 - p) + pk/2^n]$ - re-verified in the re-read; the dangling "conjectures closed" narrative is gone; and the stray "+0.99958" cell of the v3's Table 3 is gone.

3. Persisting Flaws (F1-F24)

The twenty-four entries below merge the first pass's 22 persisting flaws with the re-read's confirmations and sharpenings (items A1-A22 of the re-read ledger), plus two re-read-only entries (the $\kappa_V(a)$ notation conflation and the two-paper dependency). Line numbers refer to companion_categorical_v2.tex. Each entry preserves the full content of both passes: the first-pass description and, where the re-read sharpened it, the sharpened statement. Severity classifications are collected in chapter 6.

F1 - Three incompatible optic formalisms coexist (Def def:optic L372-382; Thm thm:filtered-colimits-optic L3206-3211; Def def:hott-optic L3618-3627).

The definition still states objects as triples (M, C, R) "Following Riley" - but Riley's objects are pairs, with the residual existential in the morphism coend. The filtered-colimits theorem uses the pair shape $F(i) = (M_i, C_i)$ with morphisms (R_{ij}, f_{ij}, g_{ij}) - Riley's actual shape, imported without a translation lemma. A third shape appears in the HoTT section: an "optic from S to A with residual R " given by a forward map $R \times^h S \rightarrow A$ and a backward map $R \times^h A \rightarrow S$, which identifies forward and backward carriers per interface (the diagonal special case), yet is presented as the generalization of the 1-categorical optic category. The composition theorem's proof (L1563-1589) imports Riley's Proposition 2.3 composition for the pair formalism while the construction (L1444) quantifies over the triple formalism; the proof also misuses monoidal language ("the tensor product is $O_2 \cdot O_1$ ", "terminality of the unit residual"). Semantic residuals in Table 1 are still English phrases rather than fiber objects. The re-read adds: the corollary "cor:unification" does not type-check - T is an endo-optic morphism, and " $T(O^*) = O^*$ " treats it as a map on objects; the theorem itself disclaims exactly this, and only $R(O): S \rightarrow S$ (the realization) has fixed points. The morphism admissibility clause "satisfying the residual-compatibility condition" is a dangling reference: the condition is stated nowhere in the paper (single occurrence, grep-verified; see also N10).

F2 - The seven maps remain generic (L1670-1690).

The instantiation is six identical residual blocks $f_1(x) = (1 - \alpha)x + \alpha s \cdot \tanh(Wx + b)$ with shared parameters ($\alpha = 0.40$, $s = 1.00$, $\rho = 0.80$) plus $f_2 = 1.15 \cdot \text{Id}$. There is no mathematical connection to RAF, RPSI, IFS, Noether, WCIG, FR, or the $n = 3$ Fisher-Rao prototype. The "unification object" contraction is about a toy map. The framing is now honest ("for an explicit seven-map instantiation"), but the semantic gap persists, and W_i, b_i are never specified - only the norms $\rho_i = 0.80$ - so even the "explicit" instantiation is incomplete (re-read sharpening, absorbed into F4).

F3 - The backward component is unanalyzed (Def realization L1723-1745; Lem lip-per-optic; Thm unconditional-banach).

The realization $R(O)(x)$ includes $\text{bwd}_1(\text{Res}_O, \text{fwd}_7 \cdots \text{fwd}_1(x))$, but every Lipschitz result bounds only the forward chain $\text{fwd}_7 \cdots \text{fwd}_1$. $\text{Lip}(R(O))$ is not the product of the $\text{Lip}(f_i)$ in general; the bwd_1 /residual Lipschitz behavior is never bounded. The contraction theorem proves the wrong map's contractivity relative to the stated definition of realization.

F4 - The self-map gap, now two-sided (L1670-1690; Prop treg-lip L1749; Prop qbound L2625).

Not only does $f_2 = 1.15 \cdot \text{Id}$ push outputs up to 1.15: each contracting f_i has an unconstrained bias, so the six contracting maps carry $[0,1]^d$ into approximately $[-0.4, 1.0]^d$, and iterating them after f_2 's range $[-0.46, 1.15]$ converges to roughly $[-1.0, 1.01]^d$. So $T([0,1]^d)$ is not contained in $[0,1]^d$ on either side, and the hypothesis "T: $X \rightarrow X$ " of the regularization propositions is never verified for the instantiation. Because W_i, b_i are unspecified, the box-invariance question cannot even be checked from the paper. (The first pass had logged the one-sided approximately-1.02 escape; the re-read sharpened it to the two-sided $[-1.0, 1.01]$. The repair - an enlarged box - is drafted in the companion repair guide.)

F5 - "Control = canonical" contradiction (Prop titer-control L2702-2714; eq:f2-expansion L1683-1685).

The control proposition says a control "replaces the RPSI optic f_2 by an expansion (determinant $1.15 > 1$)" - but the canonical instantiation already defines $f_2 = 1.15 \cdot \text{Id}$ as THE expansion optic (the abstract calls it "the expansion optic f_2 "). The control is indistinguishable from the canonical configuration. Also $\det(1.15 \cdot \text{Id}) = 1.15^d$, not 1.15. The five expansion profiles of the titer proposition (canonical, $k = 3$ axis, $k = 7$ axis, rotated) are never mapped to concrete matrices.

F6 - Envelope domination: two independent defects (Thm smooth-envelope item (e), L2969-2983; proof L3019-3026).

Form mismatch: eq:kappa-alg defines the algorithmic curvature as $(1/V_{\max}) \cdot \sup_{a \in (0, a^*]} \cdot \sup$ over unit bivectors of the positive part of the directional derivative $E'(x; F(u, v))$ - a raw positive part of a directional derivative divided by $V_{\max}$ - whereas the paper's Definition def:kv defines $\kappa_\alpha = [-D_{F(u, v)} h_\alpha]^+ / h_\alpha$, a ratio with margin, per α . Nothing connects E to $h_\alpha = D_\varphi \cdot r$. Domination gap (re-read): the proof argues " $E \geq r_q$ pointwise implies $E'(x; v) \geq r_q'(x; v)$ for every q " - but the parenthetical justifies domination only for active members (those in the argmax). For non-active surrogates the implication is false (counterexample: $E = \max(0, g)$ with $g(x) = x$ at $x = -1$; then $E'(\cdot; +1) = 0 < g' = 1$). The numerics check function-value domination, not derivative domination. Additionally the symbol a^* appears exactly once in the paper (L2972) and is never defined; the supremum over a is stray, since $F(u, v)$ as displayed carries no a -dependence. The v3 unit error re-manifests softly: the theorem dominates a curvature-like functional that is not the paper's κ_v .

F7 - The v10 indicator κ_v retired, with the pointer to the application paper (Remark fba-kappa-superseded, L1272-1278).

The paper now concedes the circularity in words: an indicator-masked gene-knockout flux-rerouting statistic "imports the model's own essentiality call into the predictor and is circular as a result" - and the definition is removed from this paper. The repair is real for the theory paper. The fate of the metric lives in the application paper: an atomic curvature measure on the parametric-FBA active-set complex, the κ^u metric, tie-break robustness, and an explicit protein-layer null. (The cross-check deliverable verifies these cross-claims against the now-provided application paper; they check out. The bibliography file itself was not provided, so the key zai2026measure cannot be resolved - see F24.)

F8 - κ_V on discrete RAF sets is still not a defined object (Prop invlim L3164-3167; Prop invlim-extended item 3, L3382-3389).

The propositions state that $\kappa_V(R_{\max})$ "computed via the filtered-colimit construction and via the operational form of Definition def:kv agrees within 10^{-9} " and speak of values "scaled by the h_α at $R_{\max}$ " on "the maximal RAF's Bregman-projected structure." But Definition def:kv is defined on (θ, x) with a connection F and margins h_α ; a discrete RAF set has none of these, and neither " h_α at $R_{\max}$ " nor "Bregman-projected structure" is a defined construction. The status is now honest (a consistency check, both sides structurally zero, not a falsifiable claim), but the object remains undefined.

F9 - The HoTT layer persists with proof-sketch status (Cor hott-fixedpoint; proof sketch L3716-3718).

The sketch still asserts "the type of contractible ∞ -groupoids is equivalent to the universe type U " - false under univalence (that type is contractible, not U -equivalent). The corollary's statement ("identified with a single term in U ") is semi-defensible, but the proof justification is false. The remark on proof status now flags (ii) the Yoneda step as informal and (iii) the univalence identification as model-dependent - partial honesty; the false equivalence remains in the sketch.

F10 - Phase III disjunction and thresholds (Def autopoiesis-phase3, L3763-3802).

Passing requires ANY of Phase I/II/III (Phase III is strictly weaker); $\varphi = 0.4$ in Phase II; tolerance $\tau = 0.30$ with $n = 2$ perturbed trajectories, no test statistic and no null model (remark on the operational status, L3804-3823). The AcCoA/ALA limit-cycle conversions to passes are claimed - but see N3 for the G/H table contradiction. Open problem 7 concedes the strictness question.

F11 - Contraction-rate-as-tolerance and the trivial holim (Remark hott-falsifiable L3849-3857; Remark hott-holim L3865-3924).

Prediction 2 is "verified ... to within the analytical tolerance" given by the Lipschitz product 0.697 - a contraction rate is not a match tolerance. The 375-grid is 1-categorical; no ∞ -categorical computation is involved. The homotopy-limit verification computes the contractibility of the path space of the interval (a simplex path space) - trivial, and "failure would have falsified" overstates what such a check can do.

F12 - The RAF-to-Zeno manufactured bound, honestly labeled (Prop raf-zeno-bound, L4108-4146).

The bound $\tau_{\text{Zeno}} \geq 1/(1 + \log_2 N)$ under the binary-cascade and gap-normalization model assumptions is now declared "part of the statement, not consequences." The verification ratio 44.9-179.6 (a vacuous margin) is still presented as confirmation, and the "direction-of-effect prediction" remains an artifact of the chosen functional form.

F13 - In-sample threshold selection, now undisclosed (Prop iJO1366-external, L4221-4239).

"At the optimal threshold $\tau^* = 0.5$ " - the threshold is tuned on the same evaluation data, with no held-out disclosure (the v3 at least documented the sweep). $\kappa = 0.835$, $\text{MCC} = 0.841$ on $n = 1638$. Also FBA-versus-FBA: the dependency ratios and the essentiality labels are both computed from the same in-silico model, so the "external validity" framing stretches ("a model no one modified" is not external data). The metabolite-level $\kappa = -0.08$ is honestly reported. $F1 = 0.8625$ is consistent with $P = 0.783$ and $R = 0.960$ (re-verified).

F14 - Persistent-homology test (Def L4168-4181; Prop L4183-4197).

Betti₀/Betti₁/Betti₂ with a 10%-of-diameter threshold. Homology does not imply contractibility (a 3-sphere passes the same checks), and homology of a time-series point cloud is not contractibility of the recovery path space. Five test cases; the threshold is arbitrary.

F15 - Network A component g unexplained (Table netA, L4052).

Row g: baseline 84.4, knockout 0.37 (never depletes: $0.37 > 0.1$), recovery 164.2 (a 2x overshoot), verdict "no." The text's catalyst-fragility explanation covers rows d and e (recovery 0.00) but not row g. The v3's misexplanation was removed but nothing replaces it for g.

F16 - The square-root embedding "unit sphere" error (Prop sqrt, L1119-1131).

$\psi_a = 2\sqrt{p_a}$ gives $|\psi| = 2$ (a radius-2 sphere); the text says "unit sphere." The Fisher-Rao distance formula $2 \cdot \arccos(\Sigma\sqrt{pq})$ is correct with the factor 2. The v3's error verbatim.

F17 - The CO(2)/CO(3) stabilizer three-claims inconsistency (Remark n3-n4, L345-370; Def struct, L324-326).

"At $n = 3$ the Fisher-Rao stabilizer is CO(2) ... At $n = 4$ the stabilizer is CO(3)" is stated flatly, versus the structure definition (SO(2)/SO(3) for the prototypes, CO(r) only with a Weyl structure, per the Fisher-Weyl remark) - and the Fisher-Rao-on-simplex point stabilizer is O($n - 1$). The load-bearing content survives: so(2) is abelian at $n = 3$, so the signature is unobservable; so(3) is non-abelian at $n \geq 4$.

F18 - The orphaned Lagrangian (eq:lagrangian, L1286-1288).

$L = E[d] + \lambda I$ is introduced as motivation for the Noether section and never used (the Noether proposition uses the geodesic Lagrangian); the bridge between the two is never built. The v3's equation (22) verbatim.

F19 - The vulnerability proposition remains rhetorical (L1076-1097).

"Adaptive systems are endangered not by ..." is elevated to a proposition, but "vulnerability" is never defined as a quantity. The proof's chain mixes a margin change with a fractional rate ($\epsilon^2 D_p h_\alpha(F_{12}) \leq \epsilon^2 \kappa_V$ per unit loop area") without unit reconciliation.

F20 - Vacuous envelope numerics (Remark smooth-envelope-numeric, L3048-3063).

"The envelope identity $E(x) = \sup_{q,q} r_q(x)$ holds pointwise, maximum error 0.0" - a definition checked against itself; plus a 5-point Danskin probe on an 8x8 toy grid. Same as the v3's remark 4.13.

F21 - "Bregman-regularized" misnomer; K never instantiated; dangling hypotheses (L1752-1755; L2628-2630).

$T_{\text{reg}} = (1 - \lambda)T + \lambda \Pi_K(T)$ is Krasnoselskii-Mann averaging with a Euclidean projection, now glossed "Krasnoselskii-Mann-averaged (Bregman-regularized)" - no Bregman divergence appears anywhere. The regularization set K is abstract: if $K = X$ the projection is the identity and the regularization is vacuous; if K is a proper subset, the narrative that the dynamics "reorganize trajectories toward the interior of K" (L2673-2676) presumes a nontrivial K that is never given. The averaging bound $(1 - \lambda)\rho + \lambda \geq \rho$ only weakens the contraction. The qbound proposition still carries never-used α_i -monotonicity and β_i -cocoercivity hypotheses, and is a near-duplicate of the treg-lip proposition.

F22 - Uncited adjacent prior art persists.

The closure criteria (Def closure-criteria) are chemical organization theory (Dittrich and Speroni di Fenizio) without citation; the designed-progression battery sits in network-expansion territory without citation; there is still no citation of the categorical-autopoiesis literature (Hirota et al.; Segura). Vereshchagin-Vitányi, Hedges, Fong, Adámek, Worrell, Bartels, Milgrom-Segal, and Clarke ARE now cited.

F23 - $\kappa_V(a)$ notation is a three-object conflation (re-read only).

Definition def:kv defines $\kappa_V(\theta, x)$ pointwise; the empirical sections use $\kappa(a) = a^2$ as a function of loop amplitude (the hierarchy claims A/D at L1393-1405; "so $\kappa_V(a) = a^2$ continues to hold" at L2044-2046); and Definition def:kappa-depth computes a loop-averaged deficit $D_V(a)$. One notation, three objects; the "operational $\kappa(a)$ " is never defined. (The application paper resolves its own version of this problem explicitly - see the cross-check deliverable, finding C6.)

F24 - Two-paper dependency (re-read; updated by the cross-check).

All zai2026measure cross-claims (the κ^H atomic curvature measure, tie-break robustness, the protein-layer null) were unverifiable from the companion file alone, and the bibliography file was not part of the upload, so reference resolution cannot be checked. [Update: the application paper source has since been provided and read in full; its cross-claims are verified in the companion deliverable "Repair Fixes and Application-Paper Cross-Check" - the dependency is now substantially resolved, though the .bib files of both papers remain absent.]

4. Issues Introduced by the Partial Repairs (V1-V7)

These seven issues were introduced (or left half-repaired) by the partial repair pass itself - places where a definition was fixed but a theorem was not, or where a repair created a new mismatch. The first pass logged them as "new issues introduced by the partial repairs"; the re-read re-verified each and sharpened V5. The strato-holonomy numerics verification that the re-read logged as item B7 is moved to chapter 7 (verified sound).

V1 - Definition (O3) versus the gluing theorem's hypothesis (L576-592 versus L612-614).

The definition was repaired to the standard gauge transform $A_j = g^{-1}{}_{ij} A_i g_{ij} + g^{-1}{}_{ij} dg_{ij}$, but the theorem's hypothesis still states the old garbled form " $g^*{}_{ij} A_j = A_i + d(\log g_{ij}) - [A_i, \log g_{ij}]$ ", while the proof (L631-635) glues "via the matching condition (O3)" - the new form. The theorem's stated hypothesis is inconsistent with both the definition and its own proof. One-line fix.

V2 - The $O(\epsilon^3)$ remainder label (eq:piecewise-F, L688-693).

The displayed exponent already contains an $O(\epsilon)$ boundary term plus $O(\epsilon^2)$ curvature; with a general (non-affine) transition the next neglected order is $O(\epsilon^2)$ (second-order Taylor of $\log g$ along the wall, BCH commutators of the two resets, and stratum cross-terms). $O(\epsilon^3)$ is correct only for affine transitions - the verified regime (the numerics use affine walls). Open problem 1 concedes the $O(\epsilon^2)$ expansion is not derived.

V3 - Stratified-Giraud descent by citation (L620-648).

"Follows from the Giraud axioms ... extended to stratified spaces" - the extension to stratified spaces is asserted at construction level. The proof-status remark (L650-664) marks this honestly: cited plus verified at construction level, coherence checked on triple intersections. Acceptable-as-marked; not a complete proof of the 2-stack property for stratified bases.

V4 - Lemma uniform-lip numerator gap (L2921-2943).

The written numerator bound omits the $\sum_j w_j$ factor; the Kraft inequality ($\sum 2^{-\ell_j} \leq 1$) does not give $\sum_j 2^{-\ell_j/\tau} \leq 1$ for $\tau \neq 1$ (and τ ranges over both sides of 1). The lemma is true via the weighted-average argument (the absolute value of a weighted sum divided by the weight sum is at most the maximum term - no Kraft needed); as written, the proof leans on a wrong inequality. The denominator bound is correct and even unnecessary under the averaging view.

V5 - Three different catalysis conditions, and the "exactly Hordijk-Steel" claim is itself inexact.

(a) The RAF definition (L384-393): catalysts drawn from the reactant pool $F \cup \bigcup_{r' \in R} \text{react}(r')$ (reactants - nonstandard), plus a condition "closed under the induced closure operator" whose closure operator is never defined (dangling). (b) The deflationary operator's Φ (L3479-3484): catalyst pool $\text{prod}(S) \cup F$ (products). (c) The Hordijk-Steel standard: catalyst pool $\text{cl}_R(F)$ (the closure). The re-read's sharpening: $\text{cl}(F \cup \text{prod}(S)) = F \cup \text{prod}(S)$ identically ($\text{prod}(S)$ already contains all products, so $F \cup \text{prod}(S)$ is S -closed), hence Φ 's reactant condition degenerates to "reactants $\in F \cup \text{prod}(S)$ " and Φ 's catalyst pool $F \cup \text{prod}(S)$ contains $\text{cl}_S(F)$. Φ therefore admits futile autocatalytic cycles ($r1: X \rightarrow Y$, $r2: Y \rightarrow X$ mutually catalyzing, nothing food-reachable) that the food-generation condition excludes. The appendix's claim (L4342-4349) of "exactly the RAF conditions of Hordijk-Steel" is false in general; equivalence holds only under well-founded (food-generated) dependency.

V6 - Stray artifacts.

An unbalanced parenthesis in the invlim-extended statement (L3369, verified by grep); a line-break hyphenation "con-structs"; figure files still named `inverse_limit_raf_*.png` (cosmetic); a double blank line at L1957-1960; the title block's "X" author placeholder; and a citealp to bhattacharya2007 (the heavy-tail Student-t reference) whose existence cannot be checked because the .bib file was not uploaded.

V7 - Two-paper dependency structure (superseded by F24 and the cross-check).

The theory paper cites the application paper for the operational instantiation (κ^H , the atomic curvature measure, tie-break robustness, the protein-layer null); the application paper was not provided at the time of the two reading passes, so all cross-paper claims were unverifiable from the file. [Status change: the application paper (journal_manuscript_v3.tex) has now been provided and cross-checked; see the companion deliverable. The repair-plan correspondence noted in chapter 1 - the repairs implementing the 2026-09-02 math-coherence plan item by item - stands confirmed.]

5. New Findings from the Re-Read (N1-N18)

These eighteen findings were NEW in the re-read - none of them appears in the first-pass ledger. Every item was verified against exact line numbers in the second pass, and the suspect computations were re-derived by hand where marked. The eight items marked as "fatal-if-refereed" feed the minimal repair set in chapter 6; the concrete fix text for each repair-set item is drafted in the companion deliverable ("Repair Fixes and Application-Paper Cross-Check").

N1 - The Ito-expansion Lévy-area normalization is inconsistent three ways (Lemma ito-expand, L2281-2344).

Statement (L2283-2287): the $O(\sigma^2)$ term is $(\sigma^2/2) \cdot X_2$ with " $\text{Var}(X_2) = 1/12$." Proof (L2298-2300): the same term is $(\sigma^2/2)$ times the loop integral of $(W_1 dW_2 - W_2 dW_1)$ "of variance $\sigma^4/12$ " - implying $\text{Var}(X_2) = 1/3$, not $1/12$. Truth: for free planar Brownian motion on $[0,1]$ (Ito), the variance of that loop integral is 1 - derived two independent ways in the re-read (Riemann sums, and the integration-by-parts identity $A + B = W_1(1)W_2(1)$ with the covariance $1/2$; both routes give 1). So the term's true variance is $\sigma^4/4$ (or $\sigma^4/16$ if X_2 is the standard half-factor Lévy area, whose variance is $1/4$). The famous " $1/12$ " is the Brownian-bridge time-integral variance, not the free-BM Lévy-area variance. Propagation: the sub-leading scale " $\sigma^2/(2\sqrt{12})$ " in the $3/2$ theorem (L2370-2371) and the figure caption term " $(v/2\sqrt{12})a$ " (L2419) inherit the wrong scale (the correct value is $\sigma^2/4$ or $\sigma^2/4$ -family depending on convention). The Monte Carlo two-term fit's c_2 would expose the factor, but its fitted value is never reported. $C_{\text{fat}} = \sqrt{(5v)/2}$ is unaffected (it comes from the $O(\sigma)$ term; the $5/4$ variance computation was re-verified correct).

N2 - Mean-versus-fluctuation conflation in the $3/2$ fatigue term.

The lemma and theorem derive $C_{\text{fat}} a^{3/2}$ as the standard deviation of a mean-zero stochastic correction ($E[H] = \pi a^2 + O(\sigma^3)$: both the $O(\sigma)$ Gaussian and the Lévy-area terms are centered). Yet Claim C (the hierarchy definition, L1398-1400) presents $H(a) = c_1 a^2 + c_2 a^{3/2}$ as a deterministic two-term scaling law, with Table n4 listing $c_2 = 0.0500$ as a "Prediction" against observed 0.0526; the stress proposition's proof (L2221-2223) calls $\mu_F = \alpha\kappa(a) + C_{\text{fat}} a^{3/2}$ "the deterministic mean" of the cumulative fatigue sum; and the Monte Carlo verification (L2389-2401) fits the standard deviation of the increment, not the increment itself. By the paper's own derivation the mean of the per-loop increment has no $a^{3/2}$ term at all; the C_{fat} term is a fluctuation scale used as a drift term across Claims C, D, and the stress test. If the intended quantity were E of the absolute increment, the constant would be $\sqrt{(2/\pi)} \cdot \sqrt{(5v)/2}$, not $\sqrt{(5v)/2}$.

N3 - The G/H cross-section contradiction (Remark netG-accolla-cycle, L3954-3975; Table network-battery, L4095-4096).

The remark claims Network G's AcCoA is converted 41/42 to 42/42 by the Phase III criterion ("the pathwise + univalence-corrected verdict is 42/42 = 100%"), but the battery table shows G at 41/42 in BOTH Phase I and Phase III; the actual one-step conversion in the table is at Network H (43/44 to 44/44). Grep-verified: "42/42" occurs exactly once in the paper (L3974). Related garbles: L3760-3761 attaches AcCoA to G and ALA to H; the "two last-remaining Phase I failures are limit-cycle oscillations" passage (L4068-4074) does not match the table (the conversions occur at H and I; none at J or K); and the cross-reference range at L4074 runs backwards in document order.

N4 - The restoration control of step (v) is silently dropped from the protocol.

The autopoiesis definition (L1209-1213, step v) requires the restoration control: all concentrations return to x^* within T steps. The operational test protocol (L4001-4014) implements only knockout-collapse plus recovery-above-threshold. The passing rows of Table netA violate the dropped control: row c recovers to 3.84 against a baseline of 0.45 (8.5x), row f recovers to 443.8 against 2.65 (167x). The "causally internal" verdicts rest on the weakened criterion; the definition and the protocol are not the same test.

N5 - The filtered-colimits-optic Step 3 is ill-formed (L3262-3275).

The displayed cocone backward map invokes a map v^C that is never defined and does not exist - in a directed colimit diagram the backward (C) components are contravariant along the arrows, so there is no $C_j \rightarrow C_j$, structure map. The extension step - "an element c -bar of C_j is the pullback of c along $C_j \rightarrow C_k \leftarrow C_j$ " - is impossible: an element of C_j cannot be pulled back to C_j along a cospan with no maps into C_j . The backward component - exactly where the mathematical content of optic colimits lives - is not validly constructed. The theorem may well be true, but this proof does not establish it. (Step 2's "representatives exist by the universal property" is also loose: they exist by the construction of Set-colimits.)

N6 - ∞ -residual: homotopy product versus homotopy pullback (Thm hott-composition, L3655-3660 vs proof sketch L3671-3673 vs verification L3885-3889).

The theorem statement says the composite residual is the homotopy PRODUCT of the residuals; the same theorem's proof sketch defines it as the homotopy PULLBACK over the action (a fiber product); and the concrete verification uses the pullback. A fiber product over a nontrivial base is not the product, and it does not reduce to the 1-categorical plain-product residual of the composition theorem. Statement, sketch, and base case do not cohere.

N7 - The geometric-phase sign slip (Cor kappa-geometric-phase, L3083-3086; proof L3101-3108).

With $A = ((V - V_{\max})/2V_{\max})d\theta$ and the deficit $\delta V = V_{\max} - V > 0$ on the loop, the loop integral of A equals $\pi(V(a) - V_{\max})/V_{\max} = -\pi\delta V/V_{\max}$. Both the corollary display and the proof's first display state $+\pi\delta V/V_{\max}$; the proof substitutes the deficit where the connection contains $(V - V_{\max})$. The magnitude (the quadratic case $\pi a^2/r^2$) is right - the proof's own Stokes computation yields the negative - so the displayed formula is the magnitude of a signed quantity with the sign silently dropped.

N8 - The Noether-current identity is garbled (L1311-1313; proof L1327-1330).

" $J_{\xi}(q, \dot{q}) = g_{\varphi}(q)(\dot{q}, \xi(q)) = \langle \nabla\varphi(q), \xi(q) \rangle \cdot \dot{q}_{\text{dual}}$ ": the left side is the pairing of $\dot{q}$ with the Hessian applied to ξ ; the right side, as typeset, is a product of a scalar with an UNDEFINED object $\dot{q}_{\text{dual}}$. The correct identity is $J_{\xi} = \langle \nabla^2\varphi(q)\dot{q}, \xi \rangle$ (equivalently $\langle \dot{q}_{\text{dual}}, \xi \rangle$ with $\dot{q}_{\text{dual}} := \nabla^2\varphi\dot{q}$, which is never defined). The proof repeats the garble. Also the affine-Bregman corollary's justification mis-attributes the Hessian equality: it comes from the affine map's vanishing second derivative, not "by the chain rule" (the chain rule gives the conjugated form).

N9 - The tightness remark uses 0.69, not 0.697 (L2667-2672).

"At $\lambda = 0.5$ the bound is 0.845" - computed from $0.5 \cdot 0.69 + 0.5$; with the theorem's product 0.697 the bound is 0.8485 (at $\lambda = 0.7$: 0.907 versus 0.909). A minor numeric inconsistency between the remark and the theorem it discusses.

N10 - The optic morphism admissibility is a dangling reference (Def def:optic, L378-379).

"... satisfying the residual-compatibility condition" - the condition is never stated (grep: single occurrence).
Consequently the paper's own optic category is never defined precisely enough to compose in; all composition facts are imported from Riley's Proposition 2.3 for a different object shape (see F1).

N11 - The Bousfield-Kan construction cited to Cisinski (L3891-3893).

"By the Bousfield-Kan construction [Ch. XI, cisinski2019]" - the homotopy-limit construction is Bousfield-Kan (1972); attributing it, with a chapter number, to Cisinski's Higher Categories and Homotopical Algebra is a wrong citation.

N12 - Claim C's $c_2 = 0.0500$ is a calibration, not a derived constant.

$C_{\text{fat}} = \sqrt{(5v)/2}$ with v a free model parameter: the Claim-C operating value 0.05 corresponds to $v = 0.002$, while the derivation's Monte Carlo uses $v = 0.1$. The FORM and the $3/2$ exponent are derived; the value is fit. Table n4 presents $c_2 = 0.0500$ as a "Prediction" alongside the genuinely derived $c_1 = \pi$ and $c_{\text{comm}} = \sqrt{2}\pi^2$.

N13 - Hol versus H notation within one theorem (eq:strat-holonomy L676; eq:piecewise-F L689).

The stratified-holonomy symbol is defined as $\text{Hol}^{\text{strat}}(\gamma)$ in one display and written $H^{\text{strat}}(\gamma_e)$ in the next - two symbols for the same object inside the same theorem.

N14 - F overload in the verification remarks.

F denotes the curvature 2-form globally (the viability-curvature definition), the scalar connection amplitude in the $\text{SO}(3)/\text{CO}(3)$ setups ($A \pm = (F/2)(x \, dy - y \, dx)T_z$, L773), and the pair ($F_+ = F_- = 2$, L917) - locally interpretable, globally ambiguous.

N15 - Claim D internal notation drift (L1402-1405).

The sufficient condition sums $(a_k(\kappa_V)_k + C_{\text{fat}} a_k^{3/2} + \eta_k)$ while the failure condition sums $(a_k \kappa(a_k) + \dots) - (\kappa_V)_k$ versus $\kappa(a_k)$ in the same claim; the dimensional coherence of the summands (amplitude times curvature plus a fluctuation scale plus a noise draw) is never established.

N16 - "Fifteen orders of magnitude" is 14.3 orders (L2011-2012).

$0.1522/7.82 \times 10^{-16} = 1.95 \times 10^{14}$, so the base-10 logarithm is 14.29.

N17 - A stray citation in the contraction battery (L2621).

"Holonomy terminology follows [misner1973]" - a functionless import from the v3 text.

N18 - The small-angle regime gets the fewest trials (Prop claimF, L1997-1998).

"50 trials per regime and 20 trials in the small-angle regime" - the regime that tests the $\sqrt{2}$ -ab prediction receives the fewest trials, and no statistical treatment of the 0.9999 mean-ratio match is given.

6. Severity Triage and the Eight-Point Repair Set

The re-read closed with a three-level severity triage of everything above, followed by the minimal repair set required for mathematical coherence. The triage is reproduced in full; the concrete drafted fix text for each of the eight repair-set items - with exact replacement wording, verification notes, and the interaction with the application paper - is the subject of the companion deliverable ("Repair Fixes and Application-Paper Cross-Check").

Fatal if refereed (math level)

- N5 - the optic-colimit Step 3 is ill-formed (undefined map, impossible element pullback).
- N1 + N2 - the Lévy-area normalization and the mean/fluctuation conflation, touching a headline "exact constant" claim.
- V5 - the catalysis-condition triple mismatch, including the false "exactly Hordijk-Steel" appendix claim.
- V1 - the gluing theorem's hypothesis contradicts both its definition and its own proof.
- F1 - the three optic formalisms plus the type-broken unification corollary.
- F6 - the envelope domination gap (form mismatch plus the active-surrogates-only proof).
- F4 + F21 - the self-map and K-instantiation gaps beneath the Banach contraction theorem.

Serious (internal consistency; fixable by localized edits)

- N3 - the G/H table contradiction.
- N4 - the silently dropped step-(v) restoration control.
- N7 - the geometric-phase sign slip.
- N8 - the Noether-current garble.
- F8 - κ_V on discrete RAFs (undefined object).
- F17 - the stabilizer three-claims inconsistency.
- F23 - the $\kappa_V(a)$ three-object conflation.
- N6 - the homotopy-product-versus-pullback mismatch.
- N12 - the calibrated (not derived) c_2 presented as a prediction.

Minor / cosmetic

- N9, N10, N13-N18, V6, F15, F20 - numeric inconsistencies, dangling references, notation drift, artifacts, and vacuous numerics.

The minimal repair set for mathematical coherence

Each item below is drafted as concrete replacement text in the companion deliverable "Repair Fixes and Application-Paper Cross-Check" (chapter 2 of that document).

- (1) Pick ONE optic formalism and restate the definition, theorem, and corollary in it (Riley pairs; define the residual-compatibility condition or use coends); delete or retype the unification corollary.
- (2) Rewrite the filtered-colimits theorem's Step 3 correctly (define the cocone's backward map via a chosen common upper bound and the composition law, then prove well-definedness) or demote the theorem to a Set-level statement.
- (3) Align the RAF definition, the deflationary operator, and the Hordijk-Steel condition (state the closure operator; add the well-foundedness hypothesis that makes Φ -fixed equal HS-RAF).
- (4) Fix the gluing theorem's hypothesis to the matching condition of definition (O3); relabel the piecewise-F remainder as $O(\epsilon^2)$.
- (5) Redo the X_2 normalization (state X_2 = half the loop integral with variance 1/4, or the full loop integral with variance 1; correct the sub-leading scale) and restate Claims C and D in fluctuation language (standard deviation, or the absolute-mean with its $\sqrt{(2/\pi)}$ constant).
- (6) Prove or hypothesis-gate $T: X \rightarrow X$ (enlarged box), instantiate K, drop the never-used cocoercivity hypotheses.
- (7) Fix the G/H table row or the AcCoA remark; implement the step-(v) restoration control or amend the definition.
- (8) Restrict the envelope domination claim to active surrogates, or prove the ratio-form version.

7. Verified-Sound Computations

Everything below was re-derived or re-checked by hand in the re-read (or verified within the repairs of chapter 2) and is sound. This list matters as much as the defect list: it delimits what a referee can be told to trust, and it protects the repair work from accidentally re-breaking correct material.

- $0.92^6 \cdot 1.15 = 0.6973$ (the contraction product).
- The Zeno mixing formula $\mu = (1 - p)/[(1 - p) + pk/2^n]$ and $\Psi(p) = \mu p + (1 - \mu)P/k$ - re-verified.
- The abelian small-loop cancellation and the numerics' regime (boundary term $O(\epsilon)$ tangential, pairs-crossing).
- Triangle area $2\epsilon^2$ with crossings at $(\pm\epsilon/2, 0)$ - re-derived.
- Pentagon area $(5/2)\epsilon^2 \sin(2\pi/5) \approx 2.38\epsilon^2$; circle $\pi\epsilon^2$; ellipse $\pi\epsilon^2/2$ - re-derived.
- $\text{Var}(G) = (5/4)\sigma^2 a^2$ with $V_c = 1/8\pi^2$, $V_s = 3/8\pi^2$, and the cross-covariance $\sigma^2 a^2/4$ - re-derived.
- $C_{\text{fat}} = \sqrt{(5v)/2}$; at $v = 0.1$ the value 0.3536; the Monte Carlo $c_1 = 0.349$ (1.5% off).
- $\mu_F = 0.027 + 0.008216 = 0.0352$.
- $c_{\text{comm}} = \sqrt{2\pi^2} = 13.96$ (4.5% error vs the measured commutator).
- Table n4 E-ratios $29 = 11.47/0.40$ and $56 = 22.22/0.40$.
- The stress-test arithmetic.
- $(K_{\text{pred}}, K_{\text{obs}}) = (29, 30)$ giving 3.4%.
- $375 = 5 \cdot 5 \cdot 3 \cdot 5$ and $152 + 219 + 4 = 375$; $15 = 16 - 1$ structure arrows.
- The extended-network R_{max} RAF-ness under Φ , spot-checked reaction by reaction.

- The appendix terminal-coalgebra proof (Tarski: greatest post-fixed point equals greatest fixed point; the Hordijk-Steel iteration matches step for step; at most $|U|$ steps; polynomial) - complete and correct as a posetal argument.
- The terminal-coalgebra numerics.
- The radial-prototype re-derivations of $D_V(a) = a^2$.
- $F1 = 0.8625$ consistent with $P = 0.783$ and $R = 0.960$.
- The strato-holonomy numerics across $U(1)/SO(3)/CO(3)$ regimes and five loop geometries: rectangle ($|H_{an}| = |H_{num}|$ to 10^{-13}), triangle, pentagon, circle, ellipse - all areas and crossing points re-derived.

8. Structural Line Map

The line map of `companion_categorical_v2.tex`, as established by the two sequential passes. All line references in chapters 2-5 resolve against this map.

Lines	Content
L1-52	Header and role declaration (repairs implemented; companion to the application paper)
L125-183	Title block, abstract (four layers), status line, table of contents
L189-283	Introduction and seven contributions
L285-478	Preliminaries: D_V , structure, optic (triple form!), RAF, CPTP, Bregman, ARD, viability-weighted curvature definitions
L482-549	SAVGS definition and the 2-categorical span
L552-940	Gluing: repaired matching condition (O3), the still-garbled theorem hypothesis, the pairs-crossing theorem, five-regime numerics
L960-1115	KKT/Fisher-minimal transport, projected differential inclusion, small-loop theorem, vulnerability proposition
L1117-1140	Square-root embedding
L1142-1278	Closure test, κ_{closure} , the fba-superseded remark
L1282-1377	Noether (repaired) and the affine-Bregman corollary
L1381-1435	Hierarchy (Claims A-D)
L1439-1649	Composition: seven optics, realization definition, typed-optic remark
L1653-1955	Lipschitz analysis, Banach contraction, Zeno contraction
L1961-2803	Verdicts: Claim F, $n = 4$ table, stress test, Lévy 3/2 (exact constant), Claim G, T-iteration battery
L2807-3117	Smooth envelope (repaired chain) and the geometric-phase corollary (repaired)
L3121-3448	Filtered colimits: componentwise proof; scale-up (16 RAFs / 21 inclusions)
L3453-3577	Terminal coalgebra (posetal, complete appendix proof) and the functorial conjecture (demoted)
L3581-3976	HoTT extension (sketch-marked), Phase III, the AcCoA remark

Lines	Content
L3983-4261	Network battery (anchors A 2/5, B 0/10; the E-K table; the RAF-Zeno bound; persistent homology; iJO1366 at $\kappa = 0.835$)
L4265-4323	Open problems (seven, honest)
L4326-4400	Appendix: terminal-coalgebra proof
L4402-4443	Declarations (AI disclosure, repository)

Table 1. Line map of companion_categorical_v2.tex (4,444 lines).